

Notes de Théorie Algébrique des Nombres

GitHub Repository: [Oxke/appunti/AlgebraSuperiore](https://github.com/Oxke/appunti/AlgebraSuperiore)

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Capitolo 1

Number fields and their rings of integers

1.1 Integrality

Definizione 1

Take $A \subseteq B$ an inclusion of domains. An element $x \in B$ is said to be **integral over** A if it is a root of a monic polynomial in $A[X]$.

The ring B is said to be **integral over** A if all elements of B are integral.

Esempio 2. The complex numbers $\overline{\mathbb{Z}}$ that are integral over $\mathbb{Z}$ are called the *algebraic integers*.

Esempio 3. Let $d \neq 0, 1$ squarefree. Then

$$\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}$$

is integral over $\mathbb{Z}$. Indeed if $x = a + b\sqrt{d}$, then $(x - a)^2 - b^2d = 0$

Esempio 4. If $\zeta = e^{\frac{2\pi i}{n}}$ then $\zeta^n = 1$, thus $\zeta \in \overline{\mathbb{Z}}$

Proposizione 5. Let A be a UFD. Let $k = \mathcal{Q}(A)$ the fraction field of A . An element $x \in K$ is integral over A if and only if $x \in A$

Dimostrazione. Write $x = \frac{p}{q}$, with $p, q \in A$ and $\text{mcd}(p, q) = 1$. Then in x is integral over A , it satisfies an equation

$$x^d + a_{d-1}x^{d-1} + \cdots + a_0 = 0 \quad a_i \in A \quad \forall i$$

Then

$$\begin{aligned} \left(\frac{p}{q}\right)^d + a_{d-1}\left(\frac{p}{q}\right)^{d-1} + \cdots + a_0 &= 0 \\ \implies p^d + a_{d-1}p^{d-1}q + \cdots + a_0q^d &= 0 \end{aligned}$$

hence $q|p^d$ but they are coprime, hence $q \in A^*$ and thus $x \in A$ □

Conversely, if $x \in A$, it is a root of $X - x$.

Corollario 6. $\mathbb{Q} \cap \overline{\mathbb{Z}} = \mathbb{Z}$

Proposizione 7. Let $A \subseteq B$ inclusion of domains. Fix $b_1, \dots, b_n \in B$ not all zero. Set $P = \sum_{i=1}^n Ab_i$ and take b such that $bP \subseteq P$.

We can write

$$bb_i = \sum_{j=1}^n a_{ij}b_j \quad a_{ij} \in A$$

Then introduce $M = (a_{ij}) \in \mathcal{M}_n(A)$ and let

$$\chi_M(T) = \det(TI_n - M)$$

then $\chi_M(b) = 0$

Osservazione. M^t

Dimostrazione. Set $N = bI_n - M$. We have

$$N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

So

$$\det N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = ({}^t\text{Conj} N) N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

Hence, since there exists i_0 with $b_{i_0} \neq 0$,

$$\det N = 0 = \chi_M(b)$$

□

Esempio 8. Let $A = \mathbb{Z}$, $B = \mathbb{C}$ If we let $b_1 = 1, b_2 = \sqrt[3]{2}, b_3 = \sqrt[3]{4}$. Then we can take $b = 1 + 3\sqrt[3]{2} - \sqrt[3]{4}$

Corollario 9. Let $A \subseteq B$ inclusion of domains. Take $x_1, \dots, x_r \in B$. Then $x_1, \dots, x_r$ are integral over A if and only if $A[x_1, \dots, x_r]$ is finitely generated as an A -module.

Dimostrazione.

$\Rightarrow$ Assume $x_1, \dots, x_r$ are integral over A . By induction, we may and do assume that $r = 1$. Take $\mu \in A[X]$ monic with $\mu(x_1) = 0$. Take $f(x_1) = x \in A[x_1]$, with $f \in A[X]$. Since μ is monic, we can write

$$f = \mu Q + R$$

with $\deg R < \deg \mu$ and $Q, R \in A[X]$. Finally, $f(x_1) = R(x_1) \in \sum_{i=1}^{\deg \mu - 1} Ax_1^i$ hence $A[x_1]$ is finitely generated.

$\Leftarrow$ Assume $A[x_1, \dots, x_r]$ is finitely generated as an A -module. Let $b_1, \dots, b_n$ be generators and set $b = x_i$. Apply the previous proposition to this setting and get a monic $\chi \in A[X]$ with $\chi(x_i) = 0$. Hence, x_i is integral over A .

□

Corollario 10. Let $A \subseteq B$ inclusion of domains. Let $\overline{A}$ be the set of elements of B that are integral over A then $\overline{A}$ is a subring of B

Dimostrazione. Take $x, y \in \overline{A}$ and consider $C = A[x, y]$. Then C is finitely generated as an A -module. But $C = A[x, y, x+y, xy]$, hence $x+y, xy \in \overline{A}$ □

In particular, $\overline{\mathbb{Z}}$ is a ring.

Corollario 11. Take $A \subseteq B \subseteq C$ a tower of inclusion of domains. Assume C is integral over b and B is integral over A . Then C is integral over A .

Dimostrazione. Take $x \in C$. Let $f \in B[X]$ monic with $f(x) = 0$ let $b_1, \dots, b_n$ be the coefficients of f and set $B' = A[b_1, \dots, b_n] \subseteq B$. Since $b_1, \dots, b_n$ are integral over A , B' is a finitely generated A -module. Since x is integral over B' , $B'[x]$ is a finitely generated B' -module. Hence, $B'[x]$ is finitely generated as A -module, so x is integral over A . $\square$

In particular, if $\overline{A} = \{x \in B : x \text{ integral over } A\}$, then $\{x \in C : x \text{ integral over } A\} = \{x \in C : x \text{ integral over } \overline{A}\}$

Definizione 12

Let $A \subseteq B \subseteq C$ domains.

1. The **integral closure** of A in B is the ring

$$\overline{A} = \{x \in B : x \text{ integral over } A\}$$

2. If $B = \mathcal{Q}(A)$, then $\overline{A}$ is called the **integral closure of A** .
3. If $\overline{A} = A$ we say that A is **integrally closed** or **normal**
4. If k is a finite extension of $\mathbb{Q}$ (i.e. a number field) then the integral closure of $\mathbb{Z}$ in k is called the **ring of integers** of k and is denoted $\mathcal{O}_k := i(K) \cap \overline{\mathbb{Z}}$ with $i : K \hookrightarrow \mathbb{C}$ injective.

Esempio 13. UFDs are integrally closed. In particular, $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$

Esempio 14. If $k = \mathbb{Q}(\sqrt{d})$ with $d \neq 0, 1$ squarefree, then $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_k$. In general (depending on the value of d), $\mathbb{Z}[\sqrt{d}]$ is not necessarily equal to $\mathcal{O}_k$. If $d \equiv 1 \pmod{4}$ then $\frac{1+\sqrt{d}}{2} \in \mathcal{O}_k \setminus \mathbb{Z}[\sqrt{d}]$

The Reference Framework (RF) is having

- A a domain with $\mathcal{Q}(A) = K$
- L/K a finite separable extension
- $B \subseteq L$ the integral closure of A in L

In particular, we are interested in the cases with $A = \mathbb{Z}$ and $K = \mathbb{Q}$

1.2 How to compute rings of integers

During this section we work with RF.

Proposizione 15. Take $x \in L$. Then $x \in B$ if and only if the minimal polynomial μ of x has coefficient in A .

Dimostrazione.

$\Rightarrow$ Assume $x \in B$. Take $f \in A[X]$ monic with $b(x) = 0$. Then $\mu|f$ in $K[X]$. So the roots $z_1, \dots, z_n$ of μ are roots of f . Hence, $z_1, \dots, z_n$ are integral over A . But the coefficients of μ are in $A[z_1, \dots, z_n]$ hence they are integral over A . But they are in K and A is integrally closed, hence they are in A and $\mu \in A[X]$

$\Leftarrow$ obvious

□

Corollario 16. Take $A = \mathbb{Z}$, $K = \mathbb{Q}$ and $L = \mathbb{Q}(\sqrt{d})$ with $d \neq 0, 1$ squarefree. Then

$$\mathcal{O}_L = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \equiv 2, 3 \pmod{4} \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] & d \equiv 1 \pmod{4} \end{cases}$$

Dimostrazione. Take $a+b\sqrt{d} \in L$, with $a, b \in \mathbb{Q}$ and $b \neq 0$. The minimal polynomial is

$$\mu = (X - a)^2 - b^2 d = X^2 - 2aX + a^2 - db^2$$

Hence $x \in \mathcal{O}_L \iff \mu \in \mathbb{Z}[X] \iff 2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. If $x \in \mathcal{O}_L$, then

$$(2a)^2 - 4(a^2 - db^2) = 4db^2 = d(2b)^2$$

Since d is squarefree, $2b \in \mathbb{Z}$. Write $a = \frac{m}{z}$, $b = \frac{n}{z}$ with $m, n \in \mathbb{Z}$. We conclude in the two cases with the thesis. □

We introduce some algebraic tools. L/K is finite separable. By the primitive element theorem we know $L = K(\alpha)$, with $\alpha \in L$ and $L = K(\alpha) \cong K[X]/(\mu)$ where μ is the minimal polynomial of α .

Take L^s a separable closure of L . $\text{Hom}_K(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_K = \text{id}_K\} \cong \{\text{roots of } \mu \text{ in } L^s\}$. In particular, $|\text{Hom}_K(L, L^s)| = [L : K]$. Coming back to the question of finding out if some elements of L are integral or not over A , we know how to compute polynomials that vanish at some $x \in L$:

$$\begin{aligned} m_x : L &\longrightarrow L \\ y &\longmapsto m_x(y) = xy \end{aligned}$$

and $\chi_{m_x}(x) = 0$

Definizione 17

$\chi_{m_x} =: \chi_x$ is called the characteristic polynomial of x .

$$\chi_x = X^n - a_{n-1}X^{n-1} + \cdots + (-1)^n a_0$$

and $a_{n-1} = \text{Tr}_{L/K}(x)$ the trace of x , $a_0 = N_{L/K}(x)$ the norm of x .

Proposizione 18. We have: μ_x minimal polynomial of x . Then

$$\chi_x = (\mu_x)^{[LK(x)]} = \prod_{\sigma \in \text{Hom}_K(L, L^s)} (X - \sigma(x))$$

and

$$\text{Tr}_{L/K}(x) = \sum_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x) \quad ; \quad N_{L/K}(x) = \prod_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x)$$

Dimostrazione.

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_{n-1} \\ 1 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 1 & \cdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \cdots & 1 & -a_0 \end{pmatrix}$$

where $\mu = X^n + a_{n-1}X^{n-1} + \dots + a_0$. Since the characteristic poly of tthis matrix is μ , we get

$$\chi_x = \mu_x^v = \mu_x^{[L:K(x)]}$$

We know:

$$\begin{aligned}\mu_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x)) \\ \chi_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x))^{[L:K(x)]}\end{aligned}$$

We have to check that, for each $\tau \in \text{Hom}_K(K(x), L^s)$, there exists exactly $[L : K(x)]$ elements $\sigma \in \text{Hom}_K(L, L^s)$ with $\sigma|_{K(x)} = \tau$.

Then, by PET, $L = K(x, y)$ with some $y \in L$. Let μ_y be the minimal polynomial of y over $K(x)$. Fixing $\tau \in \text{Hom}_K(K(x), L^s)$,

$$\text{Hom}_{K(x)}(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_{K(x)} = \tau\} \text{ has } [L : K(x)] = \deg \mu_y \text{ elements}$$

□

Corollario 19. Take $x \in L$. Then $x \in B$ if and only if $\chi_x \in A[X]$. In particular, if $x \in B$, then $\text{Tr}_{L/K}(x) \in A$ and $N_{L/K}(x) \in A$

Dimostrazione.

□

Fix $\mathbf{x} = (x_1, \dots, x_n)$ a K -basis of L . Given $x = \sum_{i=1}^n d_i x_i$, $d_i \in K$, I want to express in terms of the d_i 's whether $x \in B$ or $x \notin B$.

Lemma 20. $\mathcal{Q}(B) = L$. Better, for each $x \in L$, there exists $y \in A \setminus \{0\}$ such that $yx \in B$.

Dimostrazione. Take $\mu_x \in K[X]$ the minimal polynomial of x . Write $a_i = b_i/c_i$, with $b_i, c_i \in A$. Set $c := (c_0, \dots, c_{m-1})$ and $z := cx$. Then

$$\left(\frac{z}{c}\right)^m + a_{m-1}\left(\frac{z}{c}\right)^{m-1} + \dots + a_0 = 0$$

by multiplying everything times c^m we find $z \in B$, hence $x = \frac{z}{c}$ with $z \in B$ and $c \in A$. □

Up to multiplying $\mathbf{x}$ by some well-chosen $a \in A$, we may assume $x_1, \dots, x_n \in B$. We come back to

$$x = \sum_{i=1}^n \lambda_i x_i, \quad \lambda_i \in K$$

If $x \in B$, then $\forall i, x x_i \in B_n$ and $\text{Tr}_{L/K}(x x_i) = \sum_{j=1}^n \lambda_j \text{Tr}_{L/K}(x_i x_j) \in A$.

If we set $M_{\mathbf{x}} = (\text{Tr}_{L/K}(x_i x_j)) \in \mathcal{M}_n(A)$, then $M_{\mathbf{x}} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} \in A^n$ and hence its

determinant is in A^n .

Proposizione 21. If $d(\mathbf{x}) = \det(M_{\mathbf{x}}) = \det(\text{Tr}_{L/K}(x_i x_j))$ and $d(\mathbf{x})B \subseteq \sum_{i=1}^n A x_i$

Definizione 22: discriminant

$d\mathbf{x}$ is called the **discriminant** of $\mathbf{x}$

Computation of the discriminant

$$\begin{aligned}
d(\mathbf{x}) &= \det(\mathrm{Tr}_{L/K}(x_i x_j)) = \\
&= \det\left(\sum_{\sigma \in \mathrm{Hom}_K(L, L^s)} \sigma(x_i) \sigma(x_j)\right) = \\
&= \left(\det(\sigma(x_j))_{\sigma, j}\right)^2
\end{aligned}$$

1.2.1 Theoretical algorithm to compute rings of integer

Take $A = \mathbb{Z}$, $K = \mathbb{Q}$, L a number field and $B = \mathcal{O}_L$, then

- 1) Find a $\mathbb{Q}$ -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L contained in B e.g. $\mathbf{x} = (1, x, x^2, \dots, x^{n-1})$ with $x \in B$ a primitive element for $L/\mathbb{Q}$
- 2) Compute $d(\mathbf{x})$
- 3) Say that

$$\bigoplus_i \mathbb{Z}x_i \subseteq B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i$$

- 4) Compute representation $y_1, \dots, y_n$ of the quotient

$$\left(\frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i\right) / \left(\bigoplus_i \mathbb{Z}x_i\right)$$

Proposizione 23. *In the RF. Assume A is a PID. Then*

1. *There exists a K -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L contained in B such that*

$$B = \bigoplus_i Ax_i$$

*such a basis $\mathbf{x}$ is called an **integral basis** of B*

2. *Assume further that $A = \mathbb{Z}$. Then $d(\mathbf{x})$ is independant of the choice of an integral basis $\mathbf{x}$ and if $\mathbf{y}$ is any K -basis of L contained in B , then*

$$d(\mathbf{y}) = [B : \Gamma_{\mathbf{y}}]^2 \cdot d(\mathbf{x})$$

Osservazione. If $A = \mathbb{Z}$ and $K = \mathbb{Q}$. Say you have a $\mathbb{Q}$ -basis $\mathbf{y}$ of L contained in $\mathcal{O}_L$ such that $d(\mathbf{y})$ is squarefree. Then $\mathcal{O}_L = \bigoplus \mathbb{Z}y_i$

Dimostrazione. 1) Take $\mathbf{y}$ a K -basis of L contained in B . We know

$$\bigoplus_i Ay_i \subseteq B \subseteq \frac{1}{d(\mathbf{y})} \bigoplus_i Ay_i$$

By classication of finitely generated modules over PIDs, B is a free A -module of rank $[L : K]$. Hence, there exists $\mathbf{x} = (x_1, \dots, x_n)$ such that $B = \bigoplus_i Ax_i$. Necessarily, $\mathbf{x}$ is also a K -basis of L

- 2) Let $\mathbf{x}$ be an integral basis of $B = \mathcal{O}_L$. Let $\mathbf{y}$ be any $\mathbb{Q}$ -basis of L contained in $\mathcal{O}_L$. We know that $d(\mathbf{x})$ (resp. $d\mathbf{y}$) is the determinant of the trace form $\mathrm{Tr}_{L/K}$ on the basis $\mathbf{x}$ (resp. $\mathbf{y}$). If P is the transition matrix from $\mathbf{x}$ to $\mathbf{y}$, then $d(\mathbf{y}) = d(\mathbf{x})(\det P)^2$. By Smith's nor form theorem (théorème de la base adaptée), $|\det P| = [\bigoplus_i \mathbb{Z}x_i : \bigoplus_i \mathbb{Z}y_i] = [\mathcal{O}_L : \Gamma_{\mathbf{y}}]$. We conclude with the equivalence in the thesis. Moreover, if $\mathbf{y}$ was an integral basis, then $\Gamma_{\mathbf{y}} = \mathcal{O}_L$ and thus $d(\mathbf{y}) = d(\mathbf{x})$

□

Definizione 24

Let L be a number field. The discriminant of an integral basis of L is called **discriminant** of L and it's noted Δ_L

1.3 Algebraic properties

Rings of integers do not satisfy many properties. They are not PID generally for instance.

Definizione 25: Dedekind domains

A domain A is said to be a **Dedekind domain** if it is Noetherian, integrally closed and every non-zero prime ideal in A is maximal. Equivalently, it's a Noetherian domain with Krull dimension at most 1.

Esempio 26. PIDs are Dedekind domains. UFDs are not all Dedekind domains. For instance, in $A = \mathbb{C}[x, y]$ (x) is not maximal.

Teorema 27

Working in the Framework RF (even though separability of L/K is not needed). Assume A is a Dedekind domain. Then B is also a Dedekind domain.

Osservazione. In particular, rings of integers in number fields are Dedekind domains.

Dimostrazione. (We use the assumption of separability of L/K). We already know that B is integrally closed. We know that, if $\mathbf{x} = (x_1, \dots, x_n)$ is a K -basis of L contained in B and $B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus A x_i$ which is a finitely generated A -module, hence noetherian. It follows that B is a noetherian ring.

Take a non-zero prime ideal $\mathfrak{p}$ in B . Set $\mathfrak{q} = \mathfrak{p}^c = \mathfrak{p} \cap A$. Then $\mathfrak{q}$ is a prime ideal in A . Moreover, if one takes $x \in \mathfrak{p} \setminus \{0\}$, then x satisfies some equation

$$x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0 \quad a_i \in A, a_0 \neq 0$$

Hence

$$a_0 = -x^n - a_{n-1}x^{n-1} - \dots - a_1x \in \mathfrak{p} \cap A = \mathfrak{q}$$

thus $\mathfrak{q}$ is non-zero, hence maximal. $B/\mathfrak{p}$ is a domain which is also a finite-dimensional $(A/\mathfrak{q})$ -vector space (B is finitely generated A -module). Hence, $B/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. $\square$

1.3.1 Fractional ideals

Definizione 28

Let A be a domain with $\mathcal{Q}(A) = K$. A fractional ideal I in K is an A -submodule of K such that there exists $a \in A \setminus \{0\}$ satisfying $aI \subseteq A$

Esempio 29. If A is a PID, fractional ideals are the xA with $x \in K \setminus \{0\}$. In particular if A is not a field, K is not a fractional ideal.

Teorema 30

Let A be a Dedekind domain. All non-zero prime ideals in A are invertible.

Take $\mathfrak{p}$ a non-zero prime ideal in A . If J is a fractional ideal such that $\mathfrak{p}J \subseteq A$, then we define

$$J \subseteq \{x \in K : x\mathfrak{p} \subseteq A\} =: \mathfrak{p}^{-1}$$

Note that if $y \in \mathfrak{p} \setminus \{0\}$, then $y\mathfrak{p}^{-1} \subseteq A$, hence $\mathfrak{p}^{-1}$ is a fractional ideal.

The theorem will then follow from the following

Proposizione 31. *Let A be a Dedekind domain, $\mathfrak{p}$ a non-zero prime ideal in A and I any non-zero ideal in A and I any non-zero ideal in A . Then $I \not\subseteq \mathfrak{p}^{-1}I$. In particular, $\mathfrak{p}^{-1}\mathfrak{p} = A$*

Osservazione. In particular, $\mathfrak{p} \subsetneq \mathfrak{p}^{-1}\mathfrak{p} \subseteq A$ since $\mathfrak{p}$ is maximal, $\mathfrak{p}^{-1}\mathfrak{p} = A$

Dimostrazione. *Step 1:* We prove that every nonzero ideal $I \subseteq A$ contains a product of nonzero prime ideals. Assume by contradiction this is wrong. Consider $\mathcal{E}$ a set of nonzero ideals $I \subseteq A$ that do not contain a product of nonzero prime ideals. By assumption $\mathcal{E} \neq \emptyset$ and using Notherianity, it has a maximal element I_0 . I_0 is not prime, so we can find $a, b \in A \setminus I_0$ such that $ab \in I_0$. Then $I_0 + (a)$ and $I_0 + (b)$ both contain product of nonzero prime ideals. Hence, $(I_0 + (a))(I_0 + (b)) \subseteq I_0$ contains a product of nonzero prime ideals, which leads to a contradiction.

Step 2: We prove that $A \subsetneq \mathfrak{p}^{-1}$. $\supseteq$ is obvious. Take now $a \in \mathfrak{p} \setminus \{0\}$; by *Step 1* we can write $\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (a) \subseteq \mathfrak{p}$ and we suppose r as small as possible, with all $\mathfrak{p}_i$ prime ideals. There exists i_0 such that $\mathfrak{p}_{i_0} \subseteq \mathfrak{p}$ and we may assume $i_0 = r$. Since $\mathfrak{p}_r$ is maximal, $\mathfrak{p}_r = \mathfrak{p}$. Moreover, since r is as small as possible, we can find $b \in \mathfrak{p}_1 \dots \mathfrak{p}_{r-1} \setminus (a)$. Then $a^{-1}b \notin A$ and $\forall x \in \mathfrak{p}$, $a^{-1}bx \in a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_{r-1}\mathfrak{p} = a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq a^{-1}(a) = A$. Hence, $a^{-1}b \in \mathfrak{p}^{-1}$.

Step 3: We prove the proposition $\mathfrak{p}^{-1}I \not\supseteq I$ by contradiction. Assume $\mathfrak{p}^{-1}I = I$ and take $x \in \mathfrak{p}^{-1}$. Then $x \in \mathfrak{p}^{-1}$. Then $xI \subseteq I$ and by Notherianity of A , I is finitely generated. By a previous proposition, this allows to find a monic $\chi \in A[X]$ such that $\chi(x) = 0$. But A is integrally closed, hence $x \in A$. Hence $\mathfrak{p}^{-1} = A$ which contradicts *Step 2*. $\square$

1.3.2 Factorization of ideals

Teorema 32

Let A be a Dedekind domain. Then every nonzero ideal of A can be written as a product of nonzero prime ideals, in a unique way up to the order.

Dimostrazione. Existence: Let $\mathcal{E}$ be the family of ideals in A other than (0) and (1) that are not product of prime ideals. By contradiction, assume $\mathcal{E} \neq \emptyset$. By Notherianity, we can find a maximal element $I_0 \in \mathcal{E}$. Since $I_0 \neq A$ we can find a maximal ideal $\mathfrak{p}$ of A containing I_0 . Set $J = \mathfrak{p}^{-1}I_0$.

Then

$$i_0 \subsetneq \mathfrak{p}^{-1}I_0 \subseteq \mathfrak{p}^{-1}\mathfrak{p} = A$$

So $\mathfrak{p}^{-1}I_0$ is an ideal of A not in $\mathcal{E}$. So $\mathfrak{p}^{-1}I_0 = \mathfrak{p}_1 \dots \mathfrak{p}_r$ for some prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. We conclude $I_0 = \mathfrak{p}\mathfrak{p}^{-1}I_0 = \mathfrak{p}\mathfrak{p}_1 \dots \mathfrak{p}_r$. $\square$

1.4 Ramification

Definizione 33

In the RF with A Dedekind, take $\mathfrak{p}$ a prime ideal in A . Write the factorization of $\mathfrak{p}B = p_1^{e_1} \dots p_g^{e_g}$.

1.

Proposizione 34. *Almost all prime ideals of A are unramified in L .*

Dimostrazione. Let $\alpha \in B$ a primitive element for L/K and take μ its minimal polynomial. Set $\square$

Teorema 35

Consider the *discriminant ideal* $\mathfrak{d}_{B/A}$ of A generated by the discriminants of all K -bases of L contained in B .

A prime ideal $\mathfrak{p}$ of A is ramified in $L \iff \mathfrak{p} | \mathfrak{d}_{B/A}$

Capitolo 2

Geometry of numbers

2.1 Completions

Let $(K, |\cdot|)$ a valued field and let R be the ring of all Cauchy sequences in K . let $\mathfrak{m} \subseteq R$ be the ideal of sequences converging to 0.

Lemma 36. $\mathfrak{m}$ is a maximal ideal in R

Dimostrazione. Take $(a_n)_n \in R \setminus \mathfrak{m}$. Since $(a_n)_n \notin \mathfrak{m}$, there exists $\varepsilon > 0$ and $N \geq 0$ such that $\forall n \geq N, |a_n| \geq \varepsilon$. Now take $(b_n)_n$ a sequence such that $\forall n \geq N, a_n b_n = 1$. Then $(b_n)_n \in R$ and hence $(b_n)_n$ is an inverse for $(a_n)_n$ in $R/\mathfrak{m}$. So $(R/\mathfrak{m})$ is a field and $\mathfrak{m}$ is maximal. $\square$

Definizione 37

The **completion** of $(K, |\cdot|)$ is the field $\widehat{K} = R/\mathfrak{m}$. K is embedded in $\widehat{K}$ by sending $a \in K$ to the sequence that is constant equal to a .

Take $(a_n)_n \in R$. We know $||a_m| - |a_n|| \leq |a_m - a_n|$ so $(|a_n|)_n \subseteq \mathbb{R}$ is Cauchy. Let $|(a_n)_n|$ be its limit. If we take $(b_n)_n \in R$ with $(a_n)_n - (b_n)_n \in \mathfrak{m}$, then $||a_n| - |b_n|| \leq |a_n - b_n|$, so $(|a_n|)_n$ and $(|b_n|)_n$ have the same limit. Hence $(a_m)_n \rightarrow |(a_n)_n|$ is an absolute value of $\widehat{K}$ that extends the absolute value of K . If $(a_n)_n \in R$, then $a_n \rightarrow (a_n)_n$ in $\widehat{K}$.

Proposizione 38. $(\widehat{K}, |\cdot|)$ is complete.

Osservazione. Let $(\widehat{K}', |\cdot|')$ be another complete value field such that $|\cdot|'$ extends the absolute value of K and K is dense in $\widehat{K}'$ and $\widehat{K}'$ is complete. Then there exists a unique K -isomorphism $\sigma : \widehat{K} \rightarrow \widehat{K}'$ such that $|\cdot|' \circ \sigma = |\cdot|$.

2.1.1 Completion of archimedean valued fields

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Let $(K, |\cdot|)$ be an archimedean complete valued field. There is an isomorphism $\sigma : K \rightarrow \mathbb{R}$ or $K \rightarrow \mathbb{C}$ such that

$$|\cdot| = (|\cdot|_\sigma \circ \sigma)^s$$

for some $\mathbb{R} \ni s \geq 1$

2.1.2 Completion of nonarchimedean valued fields

Proposizione 40. *Let $(K, |\cdot|)$ be a valued field with discrete normalized absolute field. Then*

1. *The valuation on $\widehat{K}$ is still discrete normalized.*
2. *Let $\mathcal{O}$ and $\widehat{\mathcal{O}}$ valuation rings of K and $\widehat{K}$ and $\mathfrak{p}$ and $\widehat{\mathfrak{p}}$ the maximal ideals. Then*

$$\forall n \geq 1, \quad \mathcal{O}/\mathfrak{p}^n \cong \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n$$

Moreover the diagonal map

$$\begin{aligned} \partial : \widehat{\mathcal{O}} &\longrightarrow \varprojlim_{n \geq 1} \mathcal{O}/\mathfrak{p}^n \\ a &\longmapsto (a \bmod \mathfrak{p}, a \bmod \mathfrak{p}^2, \dots) \end{aligned}$$

is an isomorphism and a homeomorphism (with $\mathcal{O}/\mathfrak{p}^n$ endowed with discrete topology).

3. *Let $\mathcal{R} \subseteq \mathcal{O}$ be a system of representatives of the residue field $k := \mathcal{O}/\mathfrak{p}$ containing 0. Let π be a uniformizer of K . Then every element $x \in \widehat{\mathcal{O}}$ can be uniquely written as*

$$x = \sum_{n=0}^{\infty} a_n \pi^n$$

with $a_0, a_1, \dots \in \mathcal{R}$

Dimostrazione. 1. Take $x \in \widehat{K} \setminus \{0\}$. We can find $y \in K$ with $|x - y| < |x|$. Then $|y| = |x|$. Hence $v(\widehat{K}^\times) = v(K^\times)$

2. Consider the inclusion $i : \mathcal{O} \rightarrow \widehat{\mathcal{O}}$. We have

$$\mathfrak{p}^n = \{x \in \mathcal{O} : v(x) \geq n\} = \{x \in \widehat{\mathcal{O}} : v(x) \geq n\} \cap \mathcal{O} = \widehat{\mathfrak{p}}^n \cap \mathcal{O}$$

So i induces an injective morphism $\widehat{i} : \mathcal{O}/\mathfrak{p}^n \rightarrow \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n$. For surjectivity, take $y \in \widehat{\mathcal{O}}$. Since K is dense in $\widehat{K}$, we can find $(y_m)_m$ with values in K such that $y_m \rightarrow y$. $y + \widehat{\mathfrak{p}}^n$ is open. So $y_m \in y + \widehat{\mathfrak{p}}^n$ for $m \gg 0$. So $\widehat{i}(y_m) = \widehat{i}(y)$ for $m \gg 0$ and $\widehat{i}$ is an isomorphism.

Consider the map of the statement ∂ . Then $\ker \partial = \bigcap_{n \geq 0} \widehat{\mathfrak{p}}^n = v^{-1}(\infty) = 0$. For surjectivity, take $(\bar{a}_n)_n$ in the limit. Let a_n be a lift of $\bar{a}_n$ to $\mathcal{O}$ for each n . We have $v(a_m - a_n) \geq \min\{m, n\}$, hence $(a_n)_n$ is Cauchy. Let a be its limit in $\widehat{K}$. Since $a_n \in \mathcal{O}$ for all n , $a \in \widehat{\mathcal{O}}$. The inequality $v(a_m - a_n) \geq \min\{m, n\}$ gives $v(a - a_n) \geq n$ for each n , hence $\partial(a) = (\bar{a}_n)_n$, so ∂ is an isomorphism.

For the *homeo*, $\mathfrak{p}^n$ form a basis of open neighborhoods of 0 in $\widehat{K}$. Then

$$\partial(\mathfrak{p}^n) = \ker \left(\varprojlim_n \mathcal{O}/\mathfrak{p}^n \xrightarrow{\text{proj}} \mathcal{O}/\mathfrak{p}^n \right) = \{(\bar{0}, \dots, \bar{0}, \bar{a}_{n+1}, \dots) \in \varprojlim_n \mathcal{O}/\mathfrak{p}^n\}$$

which form a basis of open neighborhoods of 0 in the inverse limit, hence ∂ is a homeomorphism.

3. We inductively construct a sequence $(a_n)_n$ with values in $\mathcal{R}$ such that $\forall m$, $v(x - \sum_{i=0}^m a_i \pi^i) \geq m + 1$. For $m = 0$, we take $a_0 \in \mathcal{R}$ such that $a_0 \equiv x \bmod \widehat{\mathfrak{p}}$. Now assume a_m is already constructed. Then

$$v \left(x - \sum_{i=0}^m a_i \pi^i \right) \geq m + 1 \implies x - \sum_{i=0}^m a_i \pi^i = b \pi^{m+1} \quad \text{with } b \in \widehat{\mathcal{O}}$$

Choose $a_{m+1} \in \mathcal{R}$ such that $a_{m+1} \equiv b \pmod{\mathfrak{p}}$. Then

$$v\left(x - \sum_{i=0}^{m+1} a_i \pi^i\right) = v(\pi^{m+1}(b - a_{m+1})) \geq m+2$$

which finishes the induction.

Then $x = \sum_{i=0}^{\infty} a_i \pi^i$. Take $b_i \in \mathcal{R}$ such that $x = \sum_{i=0}^{\infty} b_i \pi^i$ and assume $(b_i)_i \neq (a_i)_i$. Take i_0 minimal index with $b_{i_0} \neq a_{i_0}$. Then

$$0 = (a_{i_0} - b_{i_0})\pi^{i_0} + \sum_{i \geq i_0+1} (a_i - b_i)\pi^i$$

where the first addend has valuation i_0 and the second has valuation $\geq i_0 + 1$. This is a contradiction and thus $(a_n)_n$ is unique. $\square$

Definizione 41

Given p prime, the completion of $(\mathbb{Q}, |\cdot|_{v_p})$ is denoted $\mathbb{Q}_p$ and is called the **field of p -adic numbers**. Its valuation ring is denoted $\mathbb{Z}_p$ and is called the **ring of p -adic integers**. $p\mathbb{Z}_p$ is the unique maximal ideal of $\mathbb{Z}_p$ and for all n ,

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n \mathbb{Z} \quad \frac{\mathbb{Z}_p}{p^n \mathbb{Z}_p} \cong \frac{\mathbb{Z}}{p^n \mathbb{Z}}$$

and an element $x \in \mathbb{Z}_p$ can be uniquely written as $x = \sum_{i=0}^{\infty} a_i p^i$, with $a_i \in \{0, \dots, p-1\}$

Lemma 42: Hensel

Let $(K, |\cdot|)$ be a complete nonarchimedean valued field with valuation ring $\mathcal{O}$. Take $f \in \mathcal{O}[X]$ a polynomial and $a \in \mathcal{O}$ such that $f'(a) \neq 0$ and $c = \frac{|f(a)|}{|f'(a)|^2} < 1$. Consider the sequence $(a_n)_n$ defined as

$$\begin{cases} a_0 = a \\ a_{n+1} = a_n - \frac{f(a_n)}{f'(a_n)} \end{cases}$$

This sequence is well-defined and converges to a root b of f in $\mathcal{O}$ such that $|b - a| \leq c < 1$

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